

A High-Speed and Low-Power DSP-Based TRNG for FPGA Implementations

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Abstract—This brief presents an effective way to design high-throughput and low-power True Random Number Generators (TRNGs) for Field Programmable Gate Array (FPGA)-based digital systems. The proposed design makes an unconventional usage of the Digital Signal Processing (DSP) slice embedded within the AMD-Xilinx FPGA devices to implement high jitter ring oscillators as entropy sources for efficient TRNG designs. Thanks to its wide bit-width output, several configurations can be enabled to group multiple oscillators within a single DSP slice. As a result, a TRNG designed through the proposed scheme outputs up to 4 random bits per clock cycle, thus leading to a considerably high-throughput, while exploiting an ultra-compact architecture. When implemented on the AMD-Xilinx Zynq XC7Z020 System on Chip (SoC), the new architecture achieves a throughput of 800×10^6 bit/sec and an energy consumption of only 22 pJ/bit. When compared to state-of-the-art competitors it achieves a throughput rate up to $2.6 \times$ higher and an energy consumption up to $8 \times$ lower. The new TRNG has been validated by means of the NIST SP 800-22, the NIST 800 90B and the AIS statistical tests.

Index Terms—True random number generator (TRNG), FPGA, DSP, oscillators.

I. INTRODUCTION

THE RECENT boom of the Internet of Thing (IoT) technologies has made necessary equipping our daily devices with security facilities aimed at enabling services like smart banking, cashless payments, intelligent sensors for medical purposes, smart grids and meters etc. [1]. In this context, True Random Number Generators (TRNGs) are indispensable components because they are widely employed to generate cryptographic keys, nonces and challenges in authentication protocols. As an example, TRNGs are widely employed in Post-Quantum Cryptography (PQC) algorithms,

based on elliptic curves, for the point randomization operation providing protection against side-channel analysis (SCA) attacks [2], [3]. A TRNG is a hardware system that generates a sequence of random bits by exploiting one or more dynamic entropy source, like thermal/shot noise, jitter and metastability of electronic circuits, etc. Nowadays, among the approaches known in literature to design TRNGs, those aimed at Field Programmable Gate Arrays (FPGAs) receive particular interest, since they provide an unprecedented flexibility level and suite several application fields. However, achieving simultaneously high levels of security and high throughput still represents a challenge in resources and energy constrained IoT devices. Thereby, the design of efficient fully digital TRNGs operating with limited energy budget on edge applications has become of great interest [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21].

Existing FPGA-based TRNGs typically exploit jitter accumulation as physical entropy source. In its most basic version [5], such an architecture utilizes several ring-oscillators (ROs) running “simultaneously” at different nominal frequencies to produce as many outputs, which are captured at the f_s sampling frequency and then XOR-ed together. Due to the jitter accumulated over a sampling clock period by the outputs of the ROs, the time when they switch from the high to low level (or vice versa) is not deterministic. Such an architecture is very simple, requiring just oscillators based on Look-up Table (LUT) primitives and flip-flops, but, in order to achieve a satisfying entropy level, it requires a large number of RO instances and/or post-processing encoding schemes, resulting in a resources/energy overhead. Anandakumar et al. [6] proposed the usage of ROs with different and tunable frequencies, based on programmable delay lines (PDLs) implemented with LUTs. However, this approach requires additional resources for both PDL and control logic that penalize the energy efficiency. Alternative designs, including multiphase sampler [8], multi-stage feedback [16] and latch-based circuits [12], [13], [14], [15], [16], [17], [18], also rely on LUT-based ROs as basic circuits from which to extract jitter. However, modern FPGA devices include several hardware primitives whose behavior can be explored in view of their utilization as TRNG building blocks.

This brief investigates the possibility of using DSP-based oscillators for the design of high-speed and low-power TRNGs on FPGA platforms, as an alternative to the traditional LUT-based ROs. Thanks to their internal architecture, DSP slices allow implementing multiple independent oscillators through just one resource, thus enabling different TRNG configurations depending on how the outputs are grouped into

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smaller subsets. The analysis here conducted highlights that DSPs are more effective, because of their higher intrinsic jitter. Such a property allows designing an ultra-compact architecture that complies with the requirements of the IoT applications above mentioned. When implemented on the AMD-Xilinx® Zynq XC7Z020 System on Chip (SoC), the proposed TRNG is able to produce up to 4 random bits per clock cycle using a single DSP slice, thus generating a random bit sequence with the considerable throughput of 800×10^6 bits/sec and an energy consumption of only 22pJ/bit. It requires just 13 LUTs and 54 flip-flops (FFs) and its running has been physically proved through NIST and AIS statistical tests, up to 200 MHz clock frequency.

II. BACKGROUND AND RELATED WORKS

In the past, several jitter models [15], [21], have been demonstrated to assess the entropy of the random bit sequence obtained by a TRNG architecture designed as proposed in [5]. In [15], the entropy of the random sequence obtained by a single RO was correlated to the ratio between the nominal period of the output signal of the RO and the sampling clock T_{CLK} , as described in (1), where, T_{RO} and σ_{jitter} are the RO running period and the standard deviation of the jitter accumulated over a clock period, respectively.

$$H_{min} = 1 - \frac{4}{\pi^2 \ln(2)} \cdot e^{-\frac{4\pi^2 \sigma_{jitter}^2 T_{CLK}}{T_{RO}^3}} \quad (1)$$

From (1), two fundamental design guidelines can be drawn for a RO-based TRNG: the ratios $\frac{\sigma_{jitter}}{T_{RO}}$ and $\frac{T_{CLK}}{T_{RO}}$ should be as high as possible in order to increase the entropy of the obtained random bit sequence at the parity of the number of employed ROs. Alternatively, the higher are such ratios the lower is the required number of ROs for a desired entropy, leading to lower area and power consumption. In traditional architectures, such a design criterion is not easily exploitable because both σ_{jitter} and T_{RO} are directly proportional to the number of stages within the ROs [16]. To work around this issue either the value of T_{CLK} or the number of ROs could be increased. Unfortunately, the former solution reduces the throughput, whereas the latter increases the power consumption.

To overcome the throughput limitation, designs proposed in [10], [17] combine ROs with time-to-digital converters based on one or more carry chains (CC), used as tapped delay lines, and a pulse width encoder.

TRNGs based on the multiphase sampling technique [8] consist of several LUT-based ROs. Their outputs are sampled multiple times on FFs using the outputs of the internal stages of another uncorrelated oscillator as the sampling signals.

The design technique used in [13] varies the phases of two clock signals, exploiting a digital clock manager (DCM) embedded in modern FPGA devices, to force one or more FFs to enter metastability. A TRNG designed in this way achieves a high throughput, but the usage of the DCM impacts on its energy efficiency.

The above overview on traditional architectures clearly shows that designing fast ROs having a jitter with a large standard deviation could significantly contribute to achieve high-randomness, high-throughput and low-power TRNGs. In [22], free running oscillators topologies using special

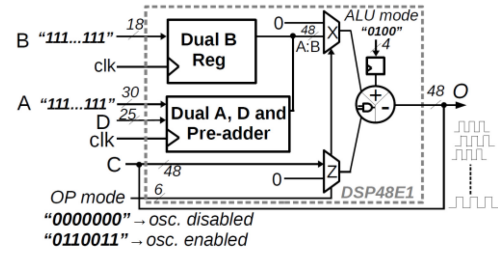


Fig. 1. Design of 48-bit DSP-based RO.

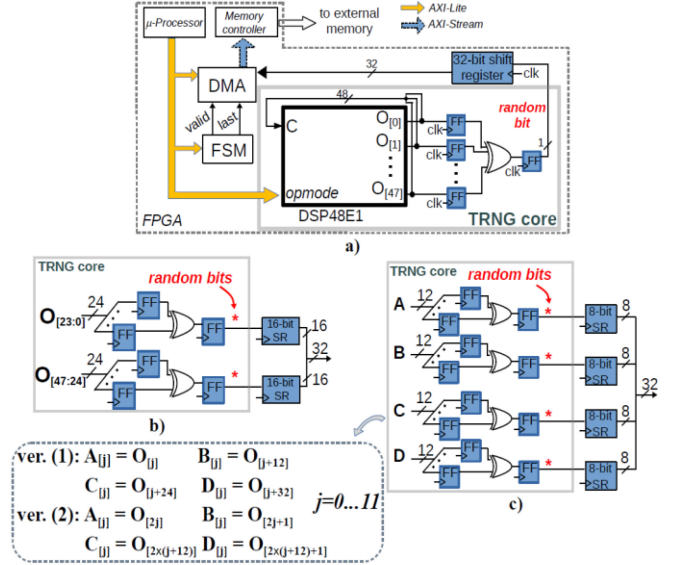


Fig. 2. The proposed TRNG architecture with throughput: (a) 1 bit/clock; (b) 2 bit/clock; (c) 4 bit/clock.

hardware primitives typically available in FPGA devices, such as fast carry chains and DSP slices, have been evaluated under the application case of Denial-of-Service (DoS) attacks identification. Our preliminary analysis on the behaviors of the DSP based oscillators convinced us to explore their exploitation in the realization of a novel TRNG architecture.

III. THE PROPOSED TRNG AND PRELIMINARY ANALYSIS

The proposed TRNG is based on the RO structure depicted in Fig. 1, which refers to the AMD-Xilinx FPGA devices equipped with the DSP48E1 and DSP48E2 slices. The latter allow to produce up to 48 output bits in parallel [25]. As highlighted in Fig. 1, the idea exploited here is to make the DSP able to act as multiple ROs by closing a feedback loop between the 48-bit output O and the input C. Then, if the DSP ALU is configured as a *bitwise XOR* through the proper *ALU_mode* settings, 48 independent oscillators are easily realized, avoiding the utilization of additional resources. Most importantly, the number of oscillators created using a single hardwired primitive provides significant benefits in terms of an increased randomness of the TRNG, without affecting its energy efficiency.

As visible in Fig. 2, the proposed TRNG includes the 48-bit DSP-based ROs whose outputs are sampled by as many FFs operating on the rising edge of the system clock. The outputs of the FFs are XOR-ed and the result, once sampled on an FF, furnishes the random bit. In its basic

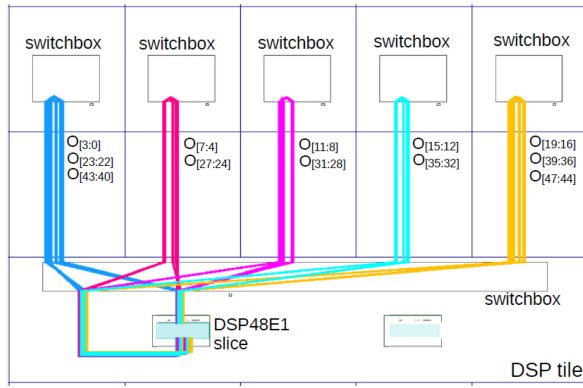


Fig. 3. The DSP tile and the routing paths of the 48 oscillators.

TABLE I
JITTER ANALYSIS OF THE 48 DSP-BASED OSCILLATORS

	$\bar{N}$	σ	$\sigma/\bar{N}$
Max	657.03×10^3	1061.1	0.195%
Min	525.14×10^3	163.4	0.026%
Mean	589.26×10^3	603.4	0.104%

TABLE II
STATISTICAL ANALYSIS OF THE ANALYZED TRNG CORES

	100MHz	1 bit/cclk	2 bit/cclk	4 bit/cclk ver (1)	4 bit/cclk ver (2)
AIS-31		✓	✓	✓	✓
Byte Entropy		7.993	7.994	7.998	7.996
200MHz					
AIS-31		✓	✓	✗ Failed tests: 1/9	✓
Byte Entropy		7.999	8.000	7.994	7.993

version (Fig. 2a), it achieves a throughput of 1 bit per clock cycle. For testing, consecutive random bits are collected into a shift register and then transferred to the off-chip DRAM memory by means of AXI-Stream transactions performed by a Direct Memory Access (DMA) module. With the aim of increasing the throughput and the energy efficiency of the TRNG, several other configurations have been analyzed, which produce more than one random bit per clock cycle, by grouping the 48 oscillators into smaller subsets. Figs. 2b-c depict the configurations outputting two and four parallel random bits, respectively. Moreover, two grouping strategies have been investigated, named *ver. (1)* and *ver. (2)*. As shown in the inset of Fig. 2c, in the former case, the groups are composed of adjacent DSP outputs; whereas in the latter, such signals are grouped in an interleaved fashion. Such an analysis was stimulated by the observation that the outputs of the DSP can be routed only through specific switchboxes, as illustrated in Fig. 3. Due to this, the feedback interconnections $O_i \rightarrow C_i$ between the i -th bits of the signals O and C forming the 48 loops are bundled into twelve groups of four consecutive signals that are spread into five switchboxes relative to the DSP tile. Therefore, to avoid detrimental effects potentially caused on randomness [23] by grouping oscillators with very similar frequencies, we adopted an interleaving grouping strategy that increases the participation of oscillators passing through distinct switchboxes in the formation of the random bits.

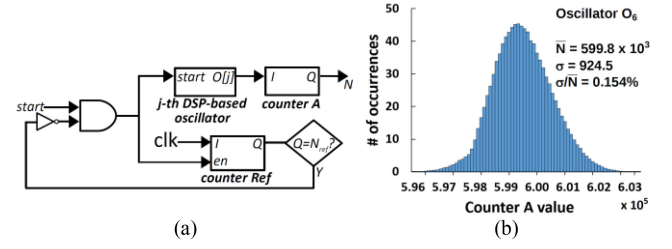
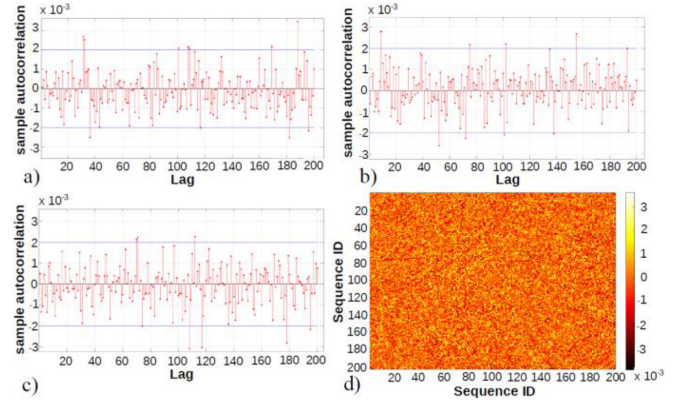


Fig. 4. (a) DSP-based oscillators jitter analysis (adapted from [9]); (b) Example of distribution of the Counter A output.

Fig. 5. (a)-(c) Autocorrelation histograms; (d) correlation matrix for 200 start-up sequences. Throughput = 4 bit/cclk, *ver. (2)*, @ $f_{\text{clk}} = 200\text{MHz}$.

A. Jitter Evaluation

In this Section, we analyze jitter measurements performed on each output bit of the DSP-based oscillator by adopting the technique described in [9] and synthetically illustrated in Fig. 4a. Such measurements were performed on an AMD-Xilinx Zynq XC7Z020 device. Table I summarizes obtained results, whereas Fig. 4b depicts an example of the distribution of the *counter A* output. There, $\bar{N}$ is the mean value of the output of *counter A* over 10^6 trials, whereas the standard deviation σ of the number of cycles of the output of the j -th oscillator, with $j = 0 \dots 47$, was evaluated within the time needed for *counter Ref* to count $N_{\text{ref}} = 10^5$ cycles of a 100MHz clock signal. As expected, each bit of the DSP-based oscillator shows a different value of σ and $\bar{N}$. It is worth noting that, compared to the LUT-based oscillators used in [9], the relative standard deviation $\sigma/\bar{N}$ and its mean value are up to $6 \times$ (0.195% vs. 0.03%) and $4 \times$ (0.104% vs. 0.025%) higher. This shows that DSP-based oscillators inherently have higher amount of jitter compared to their LUT-based counterparts, which could have beneficial effects on TRNG applications.

B. Randomness Evaluation

NIST SP 800-22 and AIS-31 statistical tests have been performed on a collection of 640 consecutive 10^6 -bit random sequences produced by the proposed TRNGs at several clock frequencies. Results summarized in Tables II and III show that, at $f_{\text{clk}} = 100\text{MHz}$, all the analyzed TRNGs pass all the statistical tests, whereas, at $f_{\text{clk}} = 200\text{MHz}$, the 4bit/cclk architecture requires the interleaving grouping to satisfy the requirements of both the test suites. As it is well known, the TRNG randomness decreases as f_{clk} increases and the number of ROs producing a single random bit decreases [11]. The

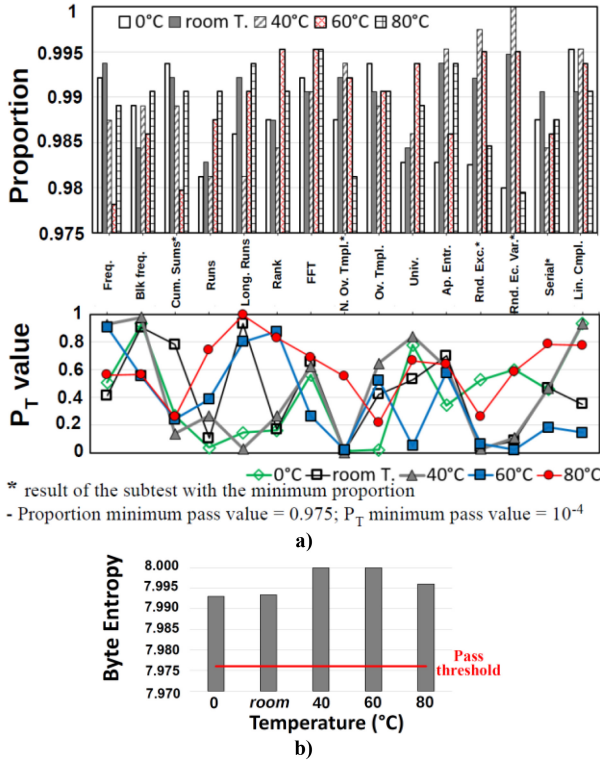


Fig. 6. (a) NIST SP800-22 and (b) AIS-31 T8 test temperature analysis. Throughput = 4 bit/clock, ver (2), @ f_{clk} = 200MHz.

TABLE III
NIST SP 800-22 TESTS RESULTS @ f_{clk} = 200MHz

Test	1 bit/clock		2 bit/clock		4 bit/clock (1)		4 bit/clock (2)	
	Prop.	P-val	Prop.	P-val	Prop.	P-val	Prop.	P-val
Frequ.	629/640	0.472	634/640	0.403	633/640	0.963	632/640	0.472
Block Freq.	635/640	0.085	635/640	0.286	630/640	0.358	630/640	0.743
Cum. Sums*	627/640	0.705	634/640	0.643	632/640	0.300	632/640	0.008
Runs	635/640	0.711	631/640	0.218	634/640	0.300	634/640	0.117
Longest Run	635/640	0.868	632/640	0.288	633/640	0.384	636/640	0.630
Rank	635/640	0.168	630/640	0.363	638/640	0.164	632/640	0.484
FFT	635/640	0.215	635/640	0.172	638/640	0.663	632/640	0.003
Non Over. Template*	627/640	0.437	629/640	0.368	623/640	0.698	627/640	0.569
Over. Template	633/640	0.607	631/640	0.941	633/640	0.078	628/640	0.102
Universal	634/640	0.215	630/640	0.927	633/640	0.406	632/640	0.353
Approx. Entropy	632/640	0.493	636/640	0.506	631/640	0.521	636/640	0.496
Random Exec.*	392/640	0.265	391/396	0.043	390/396	0.794	396/404	0.358
Random Exec. Variant*	391/399	0.578	390/396	0.284	390/396	0.629	398/404	0.242
Serial*	630/640	0.800	636/640	0.172	637/640	0.297	633/640	0.434
Linear Compl.	636/640	0.060	632/640	0.630	636/640	0.885	633/640	0.685

* result of the subtest with the minimum proportion. The failed tests are in bold.

above test confirms that the interleaving strategy improves the randomness of the proposed architecture [23], achieving a throughput up to $4 \times 200 \times 10^6$ bit/sec. When exploiting the interleaving strategy, also the TRNG of Fig. 2c successfully passed NIST 800 90B statistical tests with a minimum bit entropy value of 0.9947.

TABLE IV
COMPARISON RESULTS

	Resources					f_{clk} [MHz]	TP [10 ⁶ bit/s]	Byte [†] Entropy	Power/TP [nJ/bit]	Technology
[14]	LUT	FF	CC	DSP	DCM	450	225	7.940	-	Artix-7
[17]	32	55	17	0	0	125	12.5	7.996	0.76	Zynq-7000
[12]	4	3	0	0	0	50	0.76	7.998	32.9*	Spartan-6
[6]	528	177	0	0	0	24	6	7.994	99.6*	Spartan-3A
[23]	521	131	0	0	0	-	2.57	7.992	-	Spartan-6
[18]	15	0	0	0	0	100	100	-	0.68*	Virtex-6
[7]	128	-	-	0	0	-	2	-	-	Virtex-5
[16]	25	2	0	0	0	-	290	-	-	Virtex-6
[13]	38	121	1	1	1	300	300	7.988	0.4	Zynq-7000
[8]	24	33	0	0	0	275	275	-	0.178	Artix-7
[19]	12	10	0	0	0	200	200	8.001	-	Kintex-7
[20]	296	514	21	0	1	100	100	-	-	Zynq UltraScale+
[21]	32	64	0	0	0	400	200	7.997	-	Artix-7
new (2a)	11	51	0	1	0	200	200	7.999	0.079	Zynq-7000
new (2b)	11	52	0	1	0	200	400	8.000	0.040	
new (2c)	13	54	0	1	0	200	800	7.993	0.022	
	13	54	0	1	0	100	400	7.996	0.032	
	13	54	0	1	0	10	40	7.999	0.266	Zynq UltraScale+
	13	54	0	1	0	200	800	7.997	0.086	

[†]estimated by the AIS-31 T8 test; *derived from [7]; (2c) refers to the implementation with the interleaving strategy.

IV. TRNG CHARACTERIZATION

The behavior of the proposed TRNG at the start-up was analyzed using 200 different sequences of 10^6 bits each. Figs. 5a-c show three samples of the autocorrelation histograms and Fig. 5d depicts the correlation matrix between all the possible sequences pairs. The maximum absolute value of the autocorrelation coefficients (correlation matrix) has been found to be just 0.004 (0.003), thus proving that the proposed architecture behaves correctly at the start-up phase. Finally, its performance has been analyzed under different temperature conditions. Fig. 6a shows the proportion and P_T-value results for the NIST SP800-22 tests, whereas Fig. 6b reports the results of the AIS-31 T8 test. The proposed TRNG passed all the statistical tests for all the analyzed temperatures (the minimum proportion and P_T-value were 0.978 and 0.0012 at 60°C and 40°C, respectively). In particular, the byte entropy (test AIS-31 T8) was always higher than the minimum pass threshold, showing a maximum of 8.000 at 40°C and 60°C.

Table IV reports a comparison with previous published TRNGs in terms of resources utilization, throughput, entropy, and energy consumption per bit at the various clock frequencies. Results collected in Table IV clearly demonstrate that the design method discussed here allow to realize TRNG architectures that overcome competitors either in terms of throughput or in terms of energy consumption. The high value of f_{clk} , in conjunction with the capability to produce four random bits in parallel, makes the proposed TRNG able to achieve a throughput of 800×10^6 bits/sec, which is $2.6 \times$ higher than the one of the fastest TRNG competitor [13]. Moreover, since the proposed TRNG is based on oscillators exploiting local and dedicated interconnections within the same DSP tile, it shows an energy dissipation per bit that is about $8 \times$ lower than the energy consumption of the most energy efficient TRNG competitor [8]. Among the fastest designs, the TRNG [19], exploiting oscillators based on dual

cross-coupled XOR gates, has a LUT consumption similar to the proposed TRNG. However, its throughput is $4\times$ lower. Similar considerations arise from the comparison with [21] that, with a $2.9\times$ increasing of the LUT occupancy, achieves a throughput 50% lower than the new *ver. (2b)* implementation. The scheme of Fig. 2c has been characterized and physically tested at several running frequencies. Since the frequency of the oscillators does not depend on f_{clk} , the impact of the DSP energy consumption on the overall TRNG energy dissipation increases as the clock frequency decreases, thus reducing the energy efficiency. However, from data in Table IV it can be observed that energy consumed per bit is still several order of magnitude lower than competitors running at similar frequency. It is worth noting that, thanks to its scheme relying on dedicated interconnections and resources, the proposed TRNG can easily target other AMD-Xilinx technologies without requiring specific actions on the floor planning strategy. For the sake of demonstration, the architecture of Fig. 2c *ver. (2)* has been also implemented and characterized on the Zynq UltraScale+ XCZU9EG SoC, equipped with the DSP48E2 slices. The bitstream generated at $f_{\text{clk}} = 200\text{MHz}$ was able to pass the NIST SP 800-22, AIS-31 and NIST 800 90B statistical tests. The latter revealed a minimum bit entropy for the bitstream of 0.9949. Others results obtained using such a more advanced device are reported in Table IV that confirms the graceful behavior of the proposed TRNG in terms of low energy consumed per random bit in comparison with the competitors.

V. CONCLUSION

This brief described an efficient way to design a high-speed and low-power TRNG architecture relying on the DSP slices used as multiple ROs. The presented analysis showed that such a strategy allows realizing a class of TRNG circuits characterized by high flexibility, high throughput and an energy consumed per random bit significantly lower than most of the state-of-the-art competitors. The compact layout makes the proposed architecture easily integrable in FPGA-based designs requiring security building blocks. Several TRNG configurations have been deeply analyzed and tested with running clock frequency spanning from 10MHz to 200MHz. In the most powerful version, the proposed circuit produces 4 random bit per clock cycle, achieving a total throughput rate of 800×10^6 bit/sec and dissipates only 22pJ/bit.

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